

Hidden Runs, Visible Rows: Certified Aggregate Frontiers under Timestamp and Capacity Uncertainty

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Abstract

Public event streams often retain one row per transaction while suppressing the key that joins rows into a shared event. In ride-pooling data, this leaves trip outcomes visible but co-rider and vehicle-run membership hidden. We introduce RUNFRONTIER, a partial-identification framework that optimizes aggregate queries over every temporally ordered latent-run decomposition consistent with released rows, timestamp supports, and a declared simultaneous-capacity bound. A run need not be a clique: sequential members may form an $A-B-C$ chain even when A and C never overlap. We prove monotonicity of reachable support and fixed-estimand endpoints under timestamp-support expansion and capacity relaxation. For a fixed root and temporal span, interval connectivity and occupancy admit a consecutive-ones formulation, giving an exact integral LP pricing oracle. Dantzig–Wolfe column generation then certifies the full fixed-time master LP, while an explicit counterexample shows that the integer master remains nonintegral. Two public-data audits illustrate the geometry. Chicago provides a declared two-transaction benchmark; NYC provides second-level timestamps but unknown latent-run membership. Across six predeclared NYC audit windows, artificial timestamp uncertainty expands certified reachable support at capacities two and three in all six windows. On one exact-time cohort, column generation closes the $C = 4$ LP after generating 102 of 1,719 feasible run columns. These are conditional feasible-world statements: we do not recover partners, vehicle runs, realized capacity, or a population effect.

CCS Concepts

• Information systems → Data provenance; • Theory of computation → Mathematical optimization; • Computing methodologies → Structured outputs.

Keywords

partial identification, hidden event membership, interval graphs, column generation, ride pooling

ACM Reference Format:

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1 Introduction

Many public datasets are row-complete but relation-incomplete. A release may preserve every transaction’s timestamp, location, duration, and outcome while removing the key that says which rows belonged to the same referral episode, household visit, shared ride, or service run. Analysts can still count rows, but relational statistics—within-event dispersion, group composition, exposure, or resource utilization—depend on an unobserved partition. Selecting

one linkage or one clustering converts release uncertainty into analyst certainty and can make downstream estimates look much more precise than the public data justify.

Ride pooling makes the distinction concrete. A public row can report that a trip was successfully shared without revealing its co-riders or vehicle-run identifier. Pairwise reconstruction is already ambiguous, but high-capacity pooling is not pairwise. Three trips may satisfy $A \leftrightarrow B$ and $B \leftrightarrow C$ while $A \nleftrightarrow C$; all three can belong to one run if passengers enter and leave sequentially and simultaneous occupancy never exceeds capacity. Timestamp coarsening adds another layer: the public record may reveal only a set containing the exact interval. Treating the entire outer envelope as the realized trip is incorrect because it can create artificial concurrency. The right quantifier is existential: a latent exact timestamp completion must exist inside each released support.

We ask a deliberately different question from partner recovery:

Which aggregate conclusions hold over all latent temporal-event worlds consistent with the declared release information?

The output is a lower and upper endpoint, each accompanied by a feasible-world witness or an explicit unresolved status. The interval is conditional on the candidate universe, timestamp supports, and capacity assumption; it is not a confidence interval and does not imply that every interior scalar is attainable.

This paper makes three contributions.

- (1) **Identification.** We define core–buffer ordered-run worlds with set-valued timestamps, positive-overlap connectivity, occupancy-capacity constraints, and aggregate query functionals. We prove joint nesting in time support and capacity and define a capacity-equivalent information-loss index.
- (2) **Algorithms.** We characterize one fixed-span rooted run by segment coverage and boundary bridge rows. The resulting consecutive-ones structure gives an integral LP oracle for additive pricing. A Dantzig–Wolfe master uses this oracle to certify the full LP relaxation. We also exhibit master nonintegrality, locating the remaining difficulty in cross-run assignment.
- (3) **Evidence.** Chicago and NYC expose complementary release regimes. The audits separate hidden-linkage ambiguity, time-support uncertainty, and capacity uncertainty; they publish only aggregate endpoints and solver certificates, not reconstructed links.

The empirical findings are modest in scope but sharp within that scope. In a fixed NYC pairwise benchmark, replacing exact seconds by artificial 15-minute outer intervals increases candidate edges by 79.5% and several identified widths by 12–15%. In an ordered-run common-support comparison, most of the $C = 2$ to $C = 4$ widening occurs already at $C = 3$. Across six predeclared audit windows, artificial timestamp supports expand certified reachable run support at $C = 2$ and $C = 3$ in every window, but the amount equivalent

to a one-step capacity relaxation varies by regime. Fine temporal resolution therefore does not imply fine relational resolution.

2 Related Work

Uncertain linkage and aggregate queries. Record-linkage research represents globally compatible links as constrained matchings rather than independent pair decisions [14]. Database work has propagated linkage uncertainty to aggregate queries [10] and optimized extremal aggregate answers over candidate-constrained two-table integrations [19]. Attribute-uncertain databases likewise track value bounds through relational queries [7]. We share the possible-worlds viewpoint but study a different latent object: rows in one public event stream must be partitioned into temporally ordered groups whose total membership may exceed simultaneous capacity. The release can also expose only a support for each exact timestamp. Our estimand is an aggregate frontier over all such groupings and time completions, not a posterior over pair identities.

Ride pooling and temporal grouping. Shareability networks and dynamic trip-vehicle assignment solve a forward allocation problem when requests and candidate trips are observed [1, 13, 15]. Empirical work predicts sharing or matching outcomes [12, 18], studies pooled demand across neighborhoods [17], or constructs proxy groups by spatiotemporal clustering [16]. Our problem is inverse: the release preserves trip rows and outcomes but removes the event key. We do not select a most likely cluster. We certify what aggregate conclusions survive across every latent run decomposition admitted by declared public information.

Structured optimization and disclosure constraints. Integer programming is standard for globally coupled disclosure-control constraints [8], and bounded-width dynamic programming is standard in structured inference [6, 9]. Our algorithmic contribution is specific to ordered interval runs. For one rooted run and a fixed span, positive-overlap connectivity and occupancy constraints yield an integral interval LP. This becomes an exact pricing oracle for a Dantzig-Wolfe master. The coupling master is nevertheless nonintegral, which separates the tractable single-run subproblem from the unresolved integer decomposition problem.

Candidate restrictions and calibrated learning. Learning may sharpen a feasible-world set, but node-level sharing outcomes do not identify partner probabilities without assumptions. Matching-level conformal prediction and weakly supervised structured prediction already have well-developed foundations [2, 11]. We therefore keep learned scoring outside the central contribution of this paper. The main results are score-free and expose candidate-support, timestamp-support, and capacity assumptions separately.

3 Latent Temporal-Run Worlds

Released rows and boundary roles. Let $V = C_0 \cup B_0$ be a finite candidate universe. Core rows C_0 define the target cohort. Buffer rows B_0 lie near its temporal boundary and may belong to a run containing a core row, but need not be selected. This role distinction prevents an arbitrary observation window from forcing every nearby row into the target population.

For row i , the release supplies a timestamp support

$$\mathcal{T}_i \subseteq \{(s, e) : s < e\}.$$

Exact public time is the singleton case. A latent completion chooses $t_i = (s_i, e_i) \in \mathcal{T}_i$ for every row. We use half-open intervals $I_i = [s_i, e_i)$ and define positive overlap by $|I_i \cap I_j| > 0$.

Ordered runs. For a completed timestamp vector t , a subset $R \subseteq V$ is a feasible run at simultaneous capacity C if: (i) R contains at least two rows and at least one core; (ii) the graph on R joining positive-overlap pairs is connected; and (iii)

$$\max_u \sum_{i \in R} \mathbb{I}\{u \in I_i\} \leq C. \quad (1)$$

Condition (1) limits simultaneous occupancy, not total run cardinality. Thus a run may contain more than C rows through sequential turnover.

A world $F = (t, \Pi)$ consists of a timestamp completion and a collection of disjoint feasible runs such that every core row belongs to exactly one run and every buffer row belongs to at most one. Unselected buffers remain outside Π . Let $q(F)$ denote the number of selected buffers. For fixed q , write

$$\mathcal{F}(\mathcal{T}, C, q) = \{F : q(F) = q\}.$$

The reachable support set and its maximum are

$$Q(\mathcal{T}, C) = \{q : \mathcal{F}(\mathcal{T}, C, q) \neq \emptyset\}, \quad M(\mathcal{T}, C) = \max Q(\mathcal{T}, C).$$

Aggregate functionals. Let $H(F)$ be any run-label-invariant aggregate computable from public row attributes and the latent partition, such as mean miles among selected buffers, within-run dispersion, or run count per core. For nonempty $\mathcal{F}(\mathcal{T}, C, q)$ define

$$\underline{H}(\mathcal{T}, C, q) = \min_{F \in \mathcal{F}(\mathcal{T}, C, q)} H(F), \quad \overline{H}(\mathcal{T}, C, q) = \max_{F \in \mathcal{F}(\mathcal{T}, C, q)} H(F). \quad (2)$$

Both extrema are attained for the finite fixed-time models used below; the continuous-support implementation reports a feasible witness, a certified infeasibility result, or an unresolved status rather than coercing timeouts.

PROPOSITION 1 (SUPPORT AND CAPACITY NESTING). *Suppose $\mathcal{T}_i \subseteq \mathcal{T}_i'$ for every row and $C \leq C'$. Then, for every q ,*

$$\mathcal{F}(\mathcal{T}, C, q) \subseteq \mathcal{F}(\mathcal{T}', C', q).$$

Consequently $Q(\mathcal{T}, C) \subseteq Q(\mathcal{T}', C')$ and $M(\mathcal{T}, C) \leq M(\mathcal{T}', C')$. For fixed q and any H ,

$$\underline{H}(\mathcal{T}', C', q) \leq \underline{H}(\mathcal{T}, C, q), \quad \overline{H}(\mathcal{T}', C', q) \geq \overline{H}(\mathcal{T}, C, q).$$

The fixed- q condition matters. Comparing outcomes at each capacity's own maximum $M(\mathcal{T}, C)$ changes both the world set and the conditioning event; such ranges need not widen monotonically.

Capacity-equivalent information loss. For a fine support $\mathcal{T}$ and a coarser support $\mathcal{T}' \supseteq \mathcal{T}$, define

$$\kappa_{\mathcal{T} \rightarrow \mathcal{T}'}(C) = \min\{k \geq 0 : M(\mathcal{T}, C + k) \geq M(\mathcal{T}', C)\}. \quad (3)$$

It measures how many capacity steps under fine time are required to reach the support attainable under coarse time. When the coarse solver only certifies a feasible lower bound $L(\mathcal{T}', C) \leq M(\mathcal{T}', C)$, replacing $M(\mathcal{T}', C)$ by L yields a conservative lower bound on the comparison, not a sharp equality.

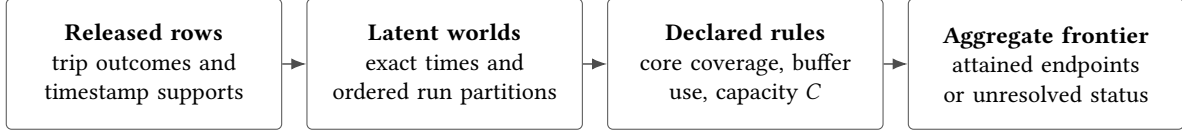


Figure 1: RUNFRONTIER never outputs a preferred partner graph. The release operator and analyst-declared support rules define a set of latent temporal worlds; optimization propagates all surviving ambiguity to a downstream aggregate.

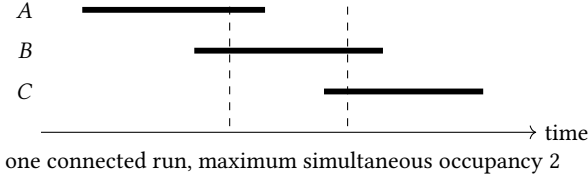


Figure 2: An ordered run need not be a clique. A overlaps B, B overlaps C, and A does not overlap C. The total run has three members but capacity two.

4 Structure-Exploiting Optimization

4.1 The declared $K = 2$ special case

If every run must contain exactly two rows, the world reduces to a core-cover matching: every core has degree one, every buffer has degree at most one, and buffer–buffer edges are excluded. Pair-additive queries become weighted matching objectives. This is the Chicago benchmark and a useful lower-complexity special case, but it cannot represent sequential high-capacity runs such as Figure 2.

4.2 A compact connectivity characterization

Fix exact intervals, a root row r , and a candidate span $[a, b]$. Let the sorted distinct endpoints inside the span induce elementary open segments. For a selected set containing r , positive-overlap connectivity with union span $[a, b]$ is equivalent to two conditions:

- (1) every elementary segment is covered by at least one selected interval;
- (2) every internal endpoint boundary is strictly crossed by at least one selected interval.

The second condition rules out components that merely touch at an endpoint. Capacity adds an upper bound of C to every segment-coverage row.

Interleave segment and boundary rows in temporal order. Each interval column then contains ones in one consecutive block. This gives the key pricing result.

THEOREM 1 (INTEGRAL ROOTED SINGLE-RUN ORACLE). *For fixed exact intervals, root r , span $[a, b]$, capacity C , and additive row weights, the linear relaxation of the segment-coverage, boundary-bridge, capacity, root, and forced-companion formulation has an integral optimum. Hence a minimum- or maximum-weight rooted run can be found by linear programming. Enumerating observed endpoint spans and, only when needed, one forced companion gives a polynomial-time exact single-run oracle.*

Proof sketch. The interleaved segment/boundary incidence matrix has the consecutive-ones property by columns. Signed duplicate

rows for lower and upper coverage bounds, plus identity rows fixing the root and companion, preserve total unimodularity. The right-hand side is integral, so every extreme point is integral. Candidate span endpoints need only be chosen from observed interval endpoints. Full details and exhaustive small-library comparisons are in the artifact.

4.3 Column generation for the full LP master

Let $\mathcal{R}_C$ be all feasible exact-time run columns. For run R , let $b_R = |R \cap B_0|$ and introduce $\lambda_R \geq 0$. The full LP relaxation for maximum reachable buffer support is

$$\max_{\lambda} \sum_{R \in \mathcal{R}_C} b_R \lambda_R \quad (4)$$

$$\text{s.t.} \quad \sum_{R \ni i} \lambda_R = 1 \quad i \in C_0, \quad (5)$$

$$\sum_{R \ni j} \lambda_R \leq 1 \quad j \in B_0. \quad (6)$$

Core equalities and buffer inequalities couple otherwise independent run columns. Given master duals, reduced cost is additive across selected rows, so Theorem 1 is an exact pricing oracle. A phase-one artificial-core master establishes feasibility. When pricing finds no improving run, the restricted master equals the full LP optimum.

PROPOSITION 2 (FULL-MASTER LP CERTIFICATE). *If every rooted pricing call is solved exactly and the terminal minimum reduced cost is nonnegative, the current restricted-master value is the optimum of Eqs. (4)–(6) over all feasible run columns.*

This certificate does not solve the full integer decomposition. The artifact contains a capacity-two instance with four cores and five buffers for which

$$\text{OPT}_{LP} = 4 > 3 = \text{OPT}_{IP}.$$

Thus integral single-run pricing does not imply an integral master. A complete integer solver requires branch-and-price, an alternative integral formulation, or a separate complexity/parameterized result.

4.4 Existential timestamp completion

For set-valued timestamps, a world is feasible when there *exists* one exact interval choice inside every support that satisfies connectivity and capacity. Replacing each support by its entire outer envelope is neither an inner nor an outer model for ordered runs: it can add bridges while also creating artificial concurrency. Our continuous mixed-integer model therefore chooses latent starts and ends jointly with run membership. Each requested support count receives one

of four statuses: certified feasible witness, proven infeasible, unresolved without incumbent, or unresolved invalid incumbent. Only the first two are treated as resolved.

5 Public-Data Audit Design

Complementary releases. Chicago’s TNP release provides a declared two-transaction shared-trip benchmark with nearest-15-minute public times and privacy-coarsened geography [3–5]. NYC’s High Volume For-Hire Vehicle release provides second-level pickup and drop-off times, shared-match indicators, Taxi Zones, trip miles, and trip time, but no public co-rider or run key. We use the official 2023 Open Data table¹ for audited small-cohort optimization and the May 2026 monthly Parquet only for current candidate-multiplicity scale.

Candidate universes. A candidate rule is deliberately visible. Chicago uses a boundary-complete core-incident temporal universe and reports radius and omitted-incidence sensitivity separately. NYC uses provider and temporal proximity; exact Taxi Zone equality is only a sensitivity screen because it can delete plausible partners. No learned score enters the headline results.

Audit levels. The experiments are designed as certification gates rather than population estimation.

- (1) A current-month NYC scale audit counts shared-match rows and candidate multiplicity for a fixed 200-row cohort.
- (2) A fixed 2023 pairwise cohort contains 38 core and 399 buffer rows and compares exact seconds with artificial 15-minute outer intervals.
- (3) Ordered-run audits use exact or set-valued times and $C \in \{2, 3, 4\}$. One common-support cohort has 8 cores; a predeclared replication panel uses 4 cores and 12 outcome-blind nearest temporal buffers in each of six windows.
- (4) A 4-core, 12-buffer exact-time cohort compares complete run-column enumeration with column generation.

The panel windows were fixed before their support frontiers were inspected. They span weekday morning, weekday evening, weekend evening, and three seasons. They are purposive, not a probability sample.

Certification and disclosure. Exact fixed-time support frontiers use complete run-column enumeration. Continuous existential-time cells use solver certificates and witness replay; unresolved counts are never classified as infeasible. Published numerical endpoints have zero reported MIP gap. Artifacts contain aggregate tables, counts, hashes, and statuses, but no raw trip identifiers, selected run columns, partner assignments, or latent timestamp witnesses.

6 Results

6.1 Fine timestamps do not identify the hidden relation

The May 2026 NYC file contains 222,149 public shared-match rows. In a fixed 200-row cohort, exact public interval overlap yields 154,708 directed core-candidate incidences over 1,042 candidate

¹Dataset u253-aew4: <https://data.cityofnewyork.us/d/u253-aew4>.

Table 1: NYC pairwise benchmark: exact seconds versus artificial 15-minute outer intervals. All displayed endpoint pairs are solver-certified with zero reported gap.

Quantity	Exact	15-min	Change
Candidate edges	5,837	10,475	+79.5%
Minimum core degree	130	260	+100.0%
Miles-gap width	12.285	13.807	+12.4%
Trip-time-gap width (min)	38.113	43.934	+15.3%
Same-dropoff-zone width	0.737	0.842	+14.3%

Table 2: Ordered-run outcome frontiers at a common support of 72 selected buffers.

C	Miles range	Width	Minutes range	Width
2	3.440–9.679	6.239	17.779–37.147	19.368
3	3.131–15.448	12.317	15.983–55.712	39.729
4	3.116–16.166	13.050	15.924–58.906	42.982

nodes; the median core degree is 786. Requiring equal pickup and drop-off Taxi Zones leaves only 44 incidences, showing why zone equality cannot be treated as a necessary partner rule.

Table 1 uses the separate fixed 2023 pairwise cohort. Exact seconds already leave broad ranges. Artificial 15-minute time coarsening adds 79.5% more candidate edges and increases the three displayed identified widths by 12.4–15.3%. Time coarsening aggravates ambiguity but does not create it.

6.2 Capacity uncertainty is front-loaded

To isolate the pure effect of capacity, Table 2 fixes the same 72 selected buffers across $C = 2, 3, 4$. The feasible sets are then nested by Proposition 1. Miles width nearly doubles from $C = 2$ to 3 but rises only 6.0% from $C = 3$ to 4. Duration width behaves similarly. In this cohort, 89.2% of the total miles widening and 86.2% of the duration widening from $C = 2$ to 4 occurs at the first relaxation. This is a cohort-specific identification result, not an estimate of realized vehicle capacity.

A different conditioning event explains an initially counterintuitive pattern. If each C is evaluated at its own maximum support, the selected-buffer composition range contracts with capacity because larger C forces the world to include more rows rather than cherry-pick extreme outcomes. Holding support fixed reverses that visual contraction exactly as nesting predicts. This separates an extensive margin—which rows can enter any feasible world—from a conditional composition margin.

6.3 Timestamp uncertainty and capacity are partially substitutable

Table 3 reports the six-window existential-time panel. Every window has a positive certified coarse-support gain at $C = 2$ and $C = 3$. The gain varies substantially, from one to seven buffers at $C = 2$. Artificial coarse $C = 2$ reaches the exact $C = 3$ maximum in two of six windows, whereas coarse $C = 3$ reaches the exact $C = 4$ maximum in all six. The latter comparison is capped by the 12-buffer

Table 3: Predeclared NYC existential-time panel. Each cell is exact maximum / coarse certified maximum / gain lower bound.

Window	$C = 2$	$C = 3$
January weekday AM	5/8/+3	9/12/+3
January weekday PM	4/8/+4	8/12/+4
January weekend PM	4/11/+7	8/12/+4
April weekday PM	7/8/+1	11/12/+1
July weekday PM	6/8/+2	10/12/+2
October weekday PM	5/8/+3	9/12/+3

Table 4: Exact-time ordered-run column generation on 4 cores and 12 buffers.

C	Full cols.	Gen. cols.	Share	Full LP	IP / restr. IP
2	54	54	100.0%	4	4 / 4
3	394	82	20.8%	8	8 / 7
4	1,719	102	5.9%	12	12 / 10

audit design, so it is evidence of support substitution within the declared panel, not universal saturation.

The correct conclusion is therefore not that 15-minute uncertainty is always “one extra seat.” Rather, timestamp information and simultaneous capacity are partially substitutable dimensions of the identified set, with a magnitude determined by local interval density and chain structure.

6.4 Exact pricing closes the LP with few columns

Table 4 compares column generation with complete enumeration on one exact-time cohort. At $C = 4$, the number of feasible run columns rises to 1,719, but 102 columns—5.9%—suffice to certify the full LP optimum. The restricted integer master is lower than the complete integer optimum at $C = 3$ and 4, matching the theoretical nonintegrality boundary. Column generation is thus a useful LP engine and lower-bound generator, not yet an exact integer solver.

7 Discussion

What is identified. The method identifies sharp or certified endpoints only relative to the declared candidate universe, timestamp supports, connectivity rule, capacity, and query. It does not identify a unique partition. The strongest empirical statement is comparative: expanding time supports or capacity cannot shrink a fixed-support feasible set, and the audits quantify how much selected support and outcome ranges move under those declared relaxations.

Algorithmic boundary. Theorem 1 makes the single-run pricing problem tractable, and Proposition 2 certifies the full LP relaxation. The exact integer master remains open. Branch-and-price is the natural next step, but branching must preserve a solvable pricing problem. A complementary goal is a hardness or fixed-parameter boundary in capacity, temporal width, or core count.

Candidate support and validation. A true run omitted before optimization cannot be recovered by endpoint certification. Future

work should pair broad public support with ground-truth or semi-synthetic recall tests, then scale the predeclared panel to many more cohorts. Point-reconstruction baselines remain useful only if evaluated by whether their downstream answers lie inside the certified frontier and how often they fail under known truth.

8 Limitations and Ethical Considerations

Neither city release exposes a public vehicle-run key. We therefore do not claim actual co-riders, partner recall outside the candidate universe, production matching logic, realized pool size, or vehicle capacity. The 15-minute NYC supports are an artificial information-loss experiment, not an assertion about the TLC observation operator. The six-window panel is too small and purposive for a citywide prevalence estimate or causal effect. A true run omitted from the declared candidate universe cannot be recovered by endpoint certification, and numerical optimality remains conditional on the recorded solver status and tolerances.

The unit is a released transaction, not a stable person. Public zones and neighborhood attributes describe context, not individual socioeconomic status or preferences. Publishing extremal partner assignments could create reidentification risk, so the artifact suppresses row identifiers and latent world witnesses and reports only aggregate results and audit metadata.

9 Generative AI Usage

Generative AI assisted drafting, code review, counterexample search, and editing. All mathematical claims, empirical outputs, citations, and artifact states remain author-verified; generated text is not treated as city evidence.

10 Conclusion

Public rows can be individually rich yet relationally incomplete. We formulate hidden shared events as ordered, bounded-occupancy interval runs with exact or set-valued timestamps and propagate their ambiguity to aggregate queries. The resulting geometry separates timestamp support, capacity, and candidate support; the algorithm separates an integral single-run pricing problem from a nonintegral cross-run master. Chicago and NYC audits show that second-level time does not identify shared-run membership and that timestamp uncertainty can produce capacity-equivalent support expansion whose magnitude varies by regime. The next decisive step is an exact integer branch-and-price method and a larger predeclared validation panel, not a more confident point linkage.

References

- [1] Javier Alonso-Mora, Samitha Samaranyake, Alex Wallar, Emilio Frazzoli, and Daniela Rus. 2017. On-Demand High-Capacity Ride-Sharing via Dynamic Trip-Vehicle Assignment. *Proceedings of the National Academy of Sciences* 114, 3 (2017), 462–467. doi:10.1073/pnas.1611675114
- [2] Maxime Cauchois, Suyash Gupta, Alnur Ali, and John C. Duchi. 2024. Predictive Inference with Weak Supervision. *Journal of Machine Learning Research* 25, 118 (2024), 1–45. <https://jmlr.org/papers/v25/23-0253.html>
- [3] City of Chicago. 2026. How Chicago Protects Privacy in TNP and Taxi Open Data. Chicago Data Portal, story 82d7-i4i2. Last edited 2026-03-27; accessed 2026-08-27. <https://data.cityofchicago.org/stories/s/82d7-i4i2>
- [4] City of Chicago. 2026. TNP Reporting Manual: Trip. Accessed 2026-08-27. <https://chicago.github.io/tnp-reporting-manual/trip/>
- [5] City of Chicago. 2026. Transportation Network Providers—Trips (2025–). Chicago Data Portal, dataset 6dvr-xwnh. Accessed 2026-08-25. <https://data.cityofchicago.org/d/6dvr-xwnh>
- [6] Rina Dechter. 1999. Bucket Elimination: A Unifying Framework for Reasoning. *Artificial Intelligence* 113, 1–2 (1999), 41–85. doi:10.1016/S0004-3702(99)00059-4
- [7] Su Feng, Boris Glavic, Aaron Huber, and Oliver A. Kennedy. 2021. Efficient Uncertainty Tracking for Complex Queries with Attribute-Level Bounds. In *Proceedings of the 2021 International Conference on Management of Data*. 528–540. doi:10.1145/3448016.3452791
- [8] Matteo Fischetti and Juan José Salazar. 2001. Solving the Cell Suppression Problem on Tabular Data with Linear Constraints. *Management Science* 47, 7 (2001), 1008–1027. doi:10.1287/mnsc.47.7.1008.9805
- [9] Eugene C. Freuder. 1990. Complexity of K -Tree Structured Constraint Satisfaction Problems. In *Proceedings of the Eighth National Conference on Artificial Intelligence*. AAAI Press, 4–9.
- [10] Ming Hua and Jian Pei. 2012. Aggregate Queries on Probabilistic Record Linkages. In *Proceedings of the 15th International Conference on Extending Database Technology*. 360–371. doi:10.1145/2247596.2247639
- [11] Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. 2018. Distribution-Free Predictive Inference for Regression. *J. Amer. Statist. Assoc.* 113, 523 (2018), 1094–1111. doi:10.1080/01621459.2017.1307116
- [12] Richard (Alex) Mucci and Gregory D. Erhardt. 2024. Understanding Ride-Hailing Sharing and Matching in Chicago Using Travel Time, Cost, and Choice Models. *Transportation Research Record* 2678, 2 (2024), 293–306. See corrigendum: 10.1177/03611981241284953. doi:10.1177/03611981231173636
- [13] Erhun Özkan and Amy R. Ward. 2020. Dynamic Matching for Real-Time Ride Sharing. *Stochastic Systems* 10, 1 (2020), 29–70. doi:10.1287/stsy.2019.0037
- [14] Mauricio Sadinle. 2017. Bayesian Estimation of Bipartite Matchings for Record Linkage. *J. Amer. Statist. Assoc.* 112, 518 (2017), 600–612. doi:10.1080/01621459.2016.1148612
- [15] Paolo Santi, Giovanni Resta, Michael Szell, Stanislav Sobolevsky, Steven H. Strogatz, and Carlo Ratti. 2014. Quantifying the Benefits of Vehicle Pooling with Shareability Networks. *Proceedings of the National Academy of Sciences* 111, 37 (2014), 13290–13294. doi:10.1073/pnas.1403657111
- [16] Elaheh Sebt and Ying Chen. 2026. Mining Hidden Ridesharing Patterns: A Data-Driven Gap Analysis of Chicago TNC Trips. *Data Science for Transportation* 8 (2026), 7. doi:10.1007/s42421-026-00149-5
- [17] Jason Soria and Amanda Stathopoulos. 2021. Investigating Socio-Spatial Differences between Solo Ridehailing and Pooled Rides in Diverse Communities. *Journal of Transport Geography* 95 (2021), 103148. doi:10.1016/j.jtrangeo.2021.103148
- [18] Morteza Taiebat, Elham Amini, and Ming Xu. 2022. Sharing Behavior in Ride-Hailing Trips: A Machine Learning Inference Approach. *Transportation Research Part D: Transport and Environment* 103 (2022), 103166. doi:10.1016/j.trd.2021.103166
- [19] Deniz Turkcapar and Sanjay Krishnan. 2023. Quantifying Uncertainty in Aggregate Queries over Integrated Datasets. arXiv:2309.05178 [cs.DB] doi:10.48550/arXiv.2309.05178

A Proof of Proposition 1

Take $F = (t, \Pi) \in \mathcal{F}(\mathcal{T}, C, q)$. Since each $t_i \in \mathcal{T}_i \subseteq \mathcal{T}'_i$, the same completion is allowed by $\mathcal{T}'$. Every run in Π has the same positive-overlap graph under the unchanged completion, and its maximum occupancy is at most $C \leq C'$. Core and buffer incidence constraints and the selected-buffer count q are unchanged. Hence $F \in \mathcal{F}(\mathcal{T}', C', q)$. Set inclusion immediately gives nesting of reachable counts and maxima. Minimizing a fixed functional over a superset can only weakly lower its optimum, and maximizing can only weakly raise it.

B Fixed-Span Connectivity Detail

Let S be a selected interval family with union span $[a, b)$. If the positive-overlap graph is connected, the union is connected and therefore covers every elementary segment. At any internal endpoint boundary, a path from an interval active immediately to its left to one active immediately to its right must contain an interval crossing the boundary strictly; endpoint touching alone is not a positive-overlap edge.

Conversely, suppose every elementary segment is covered and every internal boundary is strictly crossed. Moving from left to right, the crossing interval at each boundary overlaps intervals covering the adjacent segments with positive length. Induction joins the segment-covering intervals into one positive-overlap component. Any selected interval lies inside the covered span and overlaps the component on a segment it covers, so the full selected family is connected.

Interleaving segment and boundary rows makes each interval column consecutive. The lower-coverage, upper-capacity, and bridge constraints are signed copies of this interval matrix; root and companion fixes are identity rows. The resulting matrix is totally unimodular with integral right-hand side, proving Theorem 1.

C Master Nonintegrality Witness

At capacity two, use four cores $c_0 = [1, 2)$, $c_1 = [4, 7)$, $c_2 = [0, 2)$, and $c_3 = [1, 2)$, and five buffers $b_0 = [0, 7)$, $b_1 = [5, 6)$, $b_2 = [5, 7)$, $b_3 = [1, 2)$, and $b_4 = [2, 6)$. Complete run-column enumeration has LP value four. One fractional solution uses weights $1/2, 1, 1/2, 1/2, 1/2, 1/2$ on

$$\{c_0, c_2\}, \{c_1, b_1\}, \{c_2, b_3\}, \{c_3, b_3\}, \{c_0, b_0, b_4\}, \{c_3, b_0, b_4\}.$$

Every core has total coverage one and each buffer has use at most one. Complete integer enumeration gives value three, attained for example by $\{c_0, c_2\}$, $\{c_3, b_0, b_1\}$, and $\{c_1, b_4\}$.

D Artifact Claim Boundary

The repository contains deterministic small-instance oracles, exact fixed-time frontiers, numerical existential-time programs, solver-status validators, public-data snapshot and count contracts, and aggregate evidence manifests. Supported claims are limited to the declared candidate universes and models. No artifact output should be interpreted as recovered partner identity, a complete private run, an estimate of realized capacity, or a statement about a city's proprietary production system.